

priority-rules

Priority rules —two orders, not one

Working model for the " priority ranking" item of A1/A2/A4. It exists because the obvious reading of " priority" (rank courses by how important they are) is wrong as a grab order, and the measured data shows it.

Reproduce: `python3 src/rank_plan_priority.py` Per-course output stays local at `raw/private/plan-priority.md` (gitignored —it reveals which courses an individual chose). Aggregates are committed at `raw/measurements/plan-priority-summary.md`.

1. Why two orders

Two questions get confused:

Order	Question	What it decides
Deferability	Can this course be taken later in the year?	what may be dropped when a clash is unavoidable
Grab urgency	Will the seat still be there when I get to it?	the order in which seats are taken inside the first-come-first-served window

They are not the same order, and on the real plan they disagree:

- a course with **1 of 58 seats taken** (2% full) can sit in the " must take" group —no competition at all;
- a half-credit common course can be at **106 of 111 seats (95%)** before the window even opens.

So a single " importance" list is not enough. The tool has to emit **two outputs**: a conflict-free study plan (what to take) and a **grab order** (what to take first). Anything that emits one list will be wrong for one of the two questions.

A third, weaker order also matters:

Order	Question	Measured as
Substitutability	If this course clashes, is there another section I can move to?	number of sibling sections sharing the base course name

Substitutability is the one dimension where a clash can be resolved by moving rather than by dropping.

2. What each dimension costs to measure

Dimension	Source that carries it	Coverage on the 14 planned courses
Deferability (semester)	the two official workbooks	8 of 14 —the rest are in no official workbook at all
Grab urgency (capacity, enrolled)	the public listing	13 of 14
Substitutability (sibling sections)	the public listing	14 of 14
"Is the programme requiring it?"	no source at all	0 of 14 —human input

The last row is a design constraint, not a gap to paper over: no tool can derive the degree-programme requirement from these sources, so the ranking must accept it as an explicit input (`--must-take codes.txt`). By default the script derives a candidate must-take set from the course attribute (学科核心课 / 专业核心课 / 公共必修课 / 专业课) and labels it as a candidate in every output.

3. Finding: the most urgent courses are also the least determined

Measured on the saved plan:

Indicator	Value
planned courses	14
courses at $\geq 90\%$ of capacity	8 of 13 that have a capacity figure
courses whose semester is unknown from every source	6 of 14
of those 6, how many are at $\geq 95\%$ of capacity	5 of 6

Five of the six courses that no source can place in a semester are already at or above 95% of capacity. The courses you would have to grab first are exactly the ones whose ability to be

postponed cannot be established. That is the sharpest measured statement of the problem in this brief, and it is invisible to any check that only looks at weekday and period.

4. Tiers

Assigned by `src/rank_plan_priority.py`; the two inputs are the semester set from the official workbooks and the remaining seats from the public listing.

Tier	Meaning	Action
P0	required by the programme, offered only this semester, and at or near capacity	first seat taken, in the first seconds of the window
P1	required, offered only this semester, but not contested	must be taken; order does not matter
P2	required, contested, but also offered next semester	take early if convenient, otherwise defer
P3	required, not contested, offered in both semesters	latest in the ordering
PX	not required by the programme (elective)	ordered by urgency only, first to be dropped under a clash
PN	not offered this semester at all , yet present in the plan	flag for review —a data or planning inconsistency, not a priority
PU	deferability unknown : absent from the official workbooks	cannot be ranked on deferability; must be resolved by hand or by another source

Tier distribution on the current plan: **P0** 3, **P1** 3, **P3** 1, **PX** 1, **PU** 6.

Rule inside a tier: ascending remaining seats (a course with 0 seats left goes before one with 30).

5. Urgency distribution across the campus

From the public listing, the 347 Yuquanlu codes with a capacity figure (306 of 347 have one):

Fill ratio	Courses
full (100%)	43
90-99%	39
70-90%	39
30-70%	54
below 30%	131

More than a quarter of the campus-offered courses with a capacity figure are at or above 90% full, which is what makes the grab order a real problem rather than a theoretical one.

6. A snapshot is a snapshot

The plan's own copy of the capacity numbers and the public listing disagree, because they were captured at different times: one course reads **376 / 420** in the planner backup and **415 / 420** in the listing captured on 2026-09-27. Capacity is time-varying.

Consequence: every capacity figure must be frozen with a timestamp and a hash, exactly like the week and period fields. An unfrozen fill ratio is not evidence.

7. Open questions to confirm

1. **The must-take set.** The tier assignment above uses a candidate set derived from course attributes. The real set comes from the degree programme and has to be supplied; until then every P0/P1/P2/P3 label is provisional.
2. **Tier ordering.** Is " deferability first, urgency inside a tier" the right lexicographic order? The alternative is " urgency first" —i.e. always grab the scarcest seat, even if the course could wait. The two give different orders whenever a full elective competes with a scarce required course.
3. **Thresholds.** $\geq 90\%$ full, or ≤ 3 seats left. Both are arbitrary and should be calibrated against the observed distribution in section 5.